CDK12 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CDK12 gene fusions from the msk_impact_50k_2026 study, covering 43 fusion events. 7/14 fusions with known frame status (50.0%) were in-frame. 16/41 fusions (39.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.17342, not statistically significant at alpha=0.05). No literature benchmark is configured for CDK12.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 43 analyzed fusion events, identifying 29 distinct fusion partner genes. The most frequent partner was FBXL20, observed in 6 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 5/7 (71.4%) of in-frame fusions retained the domain, p=0.17342 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.108911.

Domain disruption

Domain-disruption analysis was skipped: CDK12 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

12% of 25 fusions with this gene as the 3' partner join at exon 1, against 1% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.0639361, corrected for testing every exon, not statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 34 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 644 aa (exon 2/3 boundary; label-permutation p=0.002997). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -84 aa from the nearest configured domain boundary at 728 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=14) and "not_retained" (n=20). For Frame_status, retained: 4/6 (66.7%, 95% CI 30.0%-90.3%); not_retained: 2/7

(28.6%, 95% CI 8.2%-64.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 21.9286 vs 44.45, p=0.102204 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for CDK12 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (19 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 4 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: CDK12 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: CDK12 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (78 row(s) in window_scan, 45 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ADCY9	2	0.046511627906976744	0.09629286886244175	None	1.0	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.28740326336468236	1
AKAP8L	1	0.023255813953488372	0.09629286886244175	None	0.9906291834002677	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2769253050196744	2
HTR1E	2	0.046511627906976744	0.09629286886244175	None	0.9250334672021419	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.27334703846508396	3
MED13	1	0.023255813953488372	0.09629286886244175	None	0.9451137884872824	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.26839116847348965	4
ODAD4	1	0.023255813953488372	0.09629286886244175	None	0.9210174029451138	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.263873096184333	5
GRB7	1	0.023255813953488372	0.09629286886244175	None	0.85809906291834	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2520759074293129	6
ERBB2	3	0.06976744186046512	0.09629286886244175	None	0.7367246764837126	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.24676007043793663	7
RECQL4	1	0.023255813953488372	0.09629286886244175	None	0.7817938420348058	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2377686785136503	8
STAT5B	4	0.09302325581395349	0.09629286886244175	None	0.5917001338688086	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.22828889893020024	9
COLEC12	1	0.023255813953488372	0.09629286886244175	None	0.6974564926372155	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.22195542550160213	10
CACNA1G	1	0.023255813953488372	0.09629286886244175	None	0.6613119143239625	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2151783170678672	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
FBXL20	6	0.13953488372093023	0.09629286886244175	None	0.4152164212405176	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.21264006327751195	12
SYNRG	1	0.023255813953488372	0.09629286886244175	None	0.6265060240963856	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2086522126501965	13
ANKFN1	1	0.023255813953488372	0.09629286886244175	None	0.6050870147255689	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.20463614839316835	14
SKAP1	1	0.023255813953488372	0.09629286886244175	None	0.5917001338688086	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.20212610823252583	15
USP6	1	0.023255813953488372	0.09629286886244175	None	0.5917001338688086	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.20212610823252583	16
CSF3	1	0.023255813953488372	0.09629286886244175	None	0.5394912985274432	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.19233695160601977	17
FAM222B	1	0.023255813953488372	0.09629286886244175	None	0.4029451137884873	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.16673454196746557	18
IKZF3	2	0.046511627906976744	0.09629286886244175	None	0.321285140562249	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.16014422722010402	19
SPON1	1	0.023255813953488372	0.09629286886244175	None	0.32797858099062915	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.15267831706786716	20
VMP1	1	0.023255813953488372	0.09629286886244175	None	0.2449799196787149	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.13711606807188326	21
SNF8	1	0.023255813953488372	0.09629286886244175	None	0.18340026773761708	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.1255698833329274	22
MAP3K3	2	0.046511627906976744	0.09629286886244175	None	None	0.4189590510404242	confidence_certainty, domain_retention, recurrence	0.12295786260268597	23
SEC31A	1	0.023255813953488372	0.09629286886244175	None	None	0.4189590510404242	confidence_certainty, domain_retention, recurrence	0.11222441000876826	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TBKBP1	1	0.023255813953488372	0.09629286886244175	None	None	0.4189590510404242	confidence_certainty, domain_retention, recurrence	0.11222441000876826	25
MTRES1	1	0.023255813953488372	0.09629286886244175	None	0.048192771084337394	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.10021847771043746	26
IGFBP4	1	0.023255813953488372	0.09629286886244175	None	0.01740294511378848	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.09444538534095955	27
FGFR4	1	0.023255813953488372	0.09629286886244175	None	0.01338688085676043	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.09369237329276679	28
RPS6KB1	1	0.023255813953488372	0.09629286886244175	None	0.0	0.4189590510404242	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.0911823331321242	29

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
80	1	13	0	20	None	0.4117647058823529	0.38535088136401713
240	2	12	0	20	None	0.1622103386809269	0.7899214689350679
246	3	11	0	20	None	0.06082887700534758	1.2158902012073491
349	4	10	0	20	None	0.021584440227703984	1.6658592098833966
603	5	9	0	20	None	0.0071948134092346605	2.1429804646030592
637	6	8	0	20	None	0.0022328731270038604	2.65113595306269
644	8	6	0	20	None	0.00016539800940769338	3.7814697215576962
700	8	6	1	19	25.333333333333332	0.0011832319134550372	2.926930125310598
703	9	5	1	19	34.2	0.0003129839254945582	3.504477966761154
750	9	5	2	18	16.2	0.0019878659658153215	2.7016129017642574
807	9	5	3	17	10.2	0.0047520594541438956	2.3231181344315144
870	9	5	4	16	7.2	0.013742615495771174	1.8619306044466175
897	9	5	5	15	5.4	0.03513617091747381	1.4542455689654064
923	9	5	8	12	2.7	0.296003611097677	0.5287029907308298
949	9	5	11	9	1.4727272727272727	0.7282366881075593	0.13772744536733278
955	9	5	12	8	1.2	1.0	-0.0
988	9	5	13	7	0.9692307692307692	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1032	10	4	13	7	1.3461538461538463	1.0	-0.0
1090	10	4	14	6	1.0714285714285714	1.0	-0.0
1121	10	4	15	5	0.8333333333333334	1.0	-0.0
1146	11	3	15	5	1.2222222222222223	1.0	-0.0
1159	11	3	16	4	0.9166666666666666	1.0	-0.0
1208	11	3	17	3	0.6470588235294118	0.6721736272626151	0.1725185310795818
1254	13	1	17	3	2.2941176470588234	0.6271778506123856	0.20260928757915825
1355	14	0	17	3	None	0.25133689839572193	0.5997437486007815
1378	14	0	18	2	None	0.5008912655971479	0.30025654135108154
1381	14	0	19	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

4	10
2	18

frame_domain_contingency_tables_by_tumor_type

stratum	table
ANSC	[1, 0], [0, 0]
BLCA	[0, 0], [0, 1]
BRCA	[0, 1], [0, 1]
CCOV	[0, 0], [0, 1]
COAD	[0, 0], [0, 2]
CUP	[0, 1], [0, 1]
GBC	[0, 1], [0, 0]
HGSOC	[0, 0], [0, 2]
HNMASC	[0, 0], [0, 1]
IDC	[1, 5], [1, 3]
PRAD	[0, 0], [0, 3]
READ	[1, 1], [0, 0]
SCCNOS	[0, 0], [1, 0]
SNUC	[0, 0], [0, 1]
STAD	[1, 0], [0, 0]

stratum	table
UMEC	[0, 0], [0, 1]
USC	[0, 1], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	34	14	12	8	0.24419795221843002

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
4	17	36	0.4722222222222222

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ADCY9	2	2	2	1
AKAP8L	1	1	1	1
ANKFN1	1	1	1	1
CACNA1G	1	1	1	1
COLEC12	1	1	1	1
CSF3	1	1	1	1
ERBB2	3	3	3	3
FAM222B	1	1	1	1
FBXL20	6	6	6	6
FGFR4	1	1	1	1
GRB7	1	1	1	1
HTR1E	2	2	2	1
IGFBP4	1	1	1	1
IKZF3	2	2	2	2
MAP3K3	2	2	2	1
MED13	1	1	1	1
MTRES1	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ODAD4	1	1	1	1
RECQL4	1	1	1	1
RPS6KB1	1	1	1	1
SEC31A	1	1	1	1
SKAP1	1	1	1	1
SNF8	1	1	1	1
SPON1	1	1	1	1
STAT5B	4	4	4	1
SYNRG	1	1	1	1
TBKBP1	1	1	1	1
USP6	1	1	1	1
VMP1	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
17	37682000	37683000	5	EVT-P-0014403-T01-IM6-87, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0057341-T01-IM6-73, EVT-P-0060111-T01-IM7-58
17	37687000	37688000	4	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0060015-T01-IM7-92
17	37619000	37620000	3	EVT-P-0036553-T01-IM6-85, EVT-P-0056622-T01-IM6-2, EVT-P-0068695-T01-IM7-93
17	37673000	37674000	3	EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0041371-T01-IM6-69
17	37627000	37628000	2	EVT-P-0039125-T01-IM6-1, EVT-P-0048104-T01-IM6-94
17	37646000	37647000	2	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56
17	37671000	37672000	2	EVT-P-0005530-T02-IM5-98, EVT-P-0046675-T01-IM6-88
17	37676000	37677000	2	EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53
17	37618000	37619000	1	EVT-P-0059578-T01-IM7-101

chromosome	bin_start	bin_end	n_events	event_ids
17	37640000	37641000	1	EVT-P-0046790-T01-IM6-83
17	37648000	37649000	1	EVT-P-0065545-T01-IM7-95
17	37650000	37651000	1	EVT-P-0002174-T01-IM3-55
17	37656000	37657000	1	EVT-P-0009835-T01-IM5-67
17	37665000	37666000	1	EVT-P-0023516-T01-IM6-78
17	37667000	37668000	1	EVT-P-0024871-T01-IM6-7
17	37668000	37669000	1	EVT-P-0043706-T01-IM6-82
17	37672000	37673000	1	EVT-P-0016507-T01-IM6-74
17	37681000	37682000	1	EVT-P-0036357-T01-IM6-84
17	37683000	37684000	1	EVT-P-0043326-T01-IM6-71

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
923	3	3	3680	False
949	3	3	1567	False
644	2	2	6628	False
1254	2	2	669	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	13	3	9	0.3333333333333333	0.06483558570881436	1.7690237700703388
three_prime	1	3	25	0.12	0.014758986704601412	2.23660136612783
three_prime	2	3	25	0.12	0.11826339114734366	0.23601435663639414
three_prime	9	3	25	0.12	0.057148328500300924	0.7715571308243299
three_prime	10	3	25	0.12	0.023909832029326474	1.6731426332334103
three_prime	13	3	25	0.12	0.0195737812551294	1.9030336922747737
three_prime	14	3	25	0.12	0.12078021557148329	0.22488130122196373
three_prime	3	2	25	0.08	0.259479126771352	0.002327184076398093
five_prime	3	1	9	0.1111111111111111	0.030010395579143187	0.6200742824075269
five_prime	5	1	9	0.1111111111111111	0.092000875417191	0.23622444603181642
five_prime	8	1	9	0.1111111111111111	0.05747660994692783	0.38403733959899283
five_prime	9	1	9	0.1111111111111111	0.02337637467855775	0.7172617646059298

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	11	1	9	0.1111111111111111	0.06453466104940636	0.34543054798258016
five_prime	14	1	9	0.1111111111111111	0.06214094216775182	0.3578875837050237
three_prime	4	1	25	0.04	0.02950429501559337	0.27817080871007377
three_prime	5	1	25	0.04	0.024675822071455928	0.33297426220678195
three_prime	7	1	25	0.04	0.1138315916178804	0.021701843969930915
three_prime	11	1	25	0.04	0.03462001422552936	0.23242548720901804
three_prime	12	1	25	0.04	0.06562893253816272	0.08789752417395037

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
444	644	200	4	30	4	10	None	0.021584440227703984	1.6658592098833966	EVT-P-0003762-T01-IM5-5, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94
500	700	200	5	29	4	10	7.6	0.13486142257489503	0.8701122635147213	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94
503	703	200	6	28	5	9	10.555555555555555	0.06082441572037855	1.215922054211928	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94, ... (6 total)
544	644	100	4	30	4	10	None	0.021584440227703984	1.6658592098833966	EVT-P-0003762-T01-IM5-5, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94
550	750	200	7	27	5	9	5.0	0.09691992885737566	1.013586913321784	EVT-P-0002174-T01-IM3-55, EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, ... (7 total)
594	644	50	4	30	4	10	None	0.021584440227703984	1.6658592098833966	EVT-P-0003762-T01-IM5-5, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94
600	700	100	5	29	4	10	7.6	0.13486142257489503	0.8701122635147213	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
603	653	50	4	30	4	10	None	0.021584440227703984	1.6658592098833966	EVT-P-0003762-T01-IM5-5, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94
603	703	100	6	28	5	9	10.555555555555555	0.06082441572037855	1.215922054211928	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94, ... (6 total)
603	803	200	7	27	5	9	5.0	0.09691992885737566	1.013586913321784	EVT-P-0002174-T01-IM3-55, EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, ... (7 total)
607	807	200	7	27	4	10	2.2666666666666666	0.40991252907270703	0.3873088071678072	EVT-P-0002174-T01-IM3-55, EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0039125-T01-IM6-1, ... (7 total)
637	737	100	5	29	4	10	7.6	0.13486142257489503	0.8701122635147213	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0065545-T01-IM7-95
637	837	200	7	27	4	10	2.2666666666666666	0.40991252907270703	0.3873088071678072	EVT-P-0002174-T01-IM3-55, EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0039125-T01-IM6-1, ... (7 total)
644	744	100	4	30	3	11	5.181818181818182	0.28303432810074175	0.5481608874418109	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0046790-T01-IM6-83, EVT-P-0065545-T01-IM7-95
644	844	200	6	28	3	11	1.5454545454545454	0.6721736272626151	0.1725185310795818	EVT-P-0002174-T01-IM3-55, EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0046790-T01-IM6-83, ... (6 total)
670	870	200	5	29	1	13	0.3076923076923077	0.3786297510206428	0.42178526412695666	EVT-P-0002174-T01-IM3-55, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0065545-T01-IM7-95
697	897	200	6	28	1	13	0.23076923076923078	0.36363115880389973	0.4393389100573004	EVT-P-0002174-T01-IM3-55, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
700	900	200	6	28	1	13	0.23076923076923078	0.36363115880389973	0.4393389100573004	EVT-P-0002174-T01-IM3-55, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (6 total)
703	903	200	5	29	1	13	0.3076923076923077	0.3786297510206428	0.42178526412695666	EVT-P-0002174-T01-IM3-55, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, EVT-P-0065545-T01-IM7-95
723	923	200	7	27	0	14	0.0	0.026212279835586775	1.5814952042744805	EVT-P-0002174-T01-IM3-55, EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (7 total)
749	949	200	10	24	0	14	0.0	0.0017219568583829524	2.7639777334868523	EVT-P-0002174-T01-IM3-55, EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, ... (10 total)
750	950	200	10	24	0	14	0.0	0.0017219568583829524	2.7639777334868523	EVT-P-0002174-T01-IM3-55, EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, ... (10 total)
755	955	200	10	24	0	14	0.0	0.0017219568583829524	2.7639777334868523	EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, ... (10 total)
788	988	200	11	23	0	14	0.0	0.0006583204286534785	3.1815626679007503	EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, ... (11 total)
807	1007	200	11	23	0	14	0.0	0.0006583204286534785	3.1815626679007503	EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, ... (11 total)
823	923	100	5	29	0	14	0.0	0.06291328848254843	1.2012576136148874	EVT-P-0005530-T02-IM5-98, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, EVT-P-0043706-T01-IM6-82, EVT-P-0046675-T01-IM6-88

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
832	1032	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (11 total)
849	949	100	8	26	0	14	0.0	0.010884048229464706	1.963209542439507	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (8 total)
855	955	100	9	25	0	14	0.0	0.004385443124565024	2.3579865170458922	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (9 total)
870	970	100	9	25	0	14	0.0	0.004385443124565024	2.3579865170458922	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (9 total)
870	1070	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (11 total)
873	923	50	4	30	0	14	0.0	0.1260565809901673	0.8994344763412977	EVT-P-0005530-T02-IM5-98, EVT-P-0024871-T01-IM6-7, EVT-P-0043706-T01-IM6-82, EVT-P-0046675-T01-IM6-88
888	988	100	9	25	0	14	0.0	0.004385443124565024	2.3579865170458922	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0024871-T01-IM6-7, EVT-P-0037167-T01-IM6-81, ... (9 total)
890	1090	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0024871-T01-IM6-7, EVT-P-0036357-T01-IM6-84, ... (11 total)
897	947	50	4	30	0	14	0.0	0.1260565809901673	0.8994344763412977	EVT-P-0005530-T02-IM5-98, EVT-P-0024871-T01-IM6-7, EVT-P-0043706-T01-IM6-82, EVT-P-0046675-T01-IM6-88

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
897	997	100	9	25	0	14	0.0	0.004385443124565024	2.3579865170458922	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0024871-T01-IM6-7, EVT-P-0037167-T01-IM6-81, ... (9 total)
897	1097	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0024871-T01-IM6-7, EVT-P-0036357-T01-IM6-84, ... (11 total)
899	949	50	6	28	0	14	0.0	0.031052774026993746	1.5078995970718483	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0041371-T01-IM6-69, EVT-P-0043706-T01-IM6-82, ... (6 total)
905	955	50	7	27	0	14	0.0	0.026212279835586775	1.5814952042744805	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0041371-T01-IM6-69, ... (7 total)
921	1121	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0014403-T01-IM6-87, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0036357-T01-IM6-84, ... (11 total)
923	973	50	7	27	0	14	0.0	0.026212279835586775	1.5814952042744805	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0041371-T01-IM6-69, ... (7 total)
923	1023	100	8	26	0	14	0.0	0.010884048229464706	1.963209542439507	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, ... (8 total)
923	1123	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0014403-T01-IM6-87, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0036357-T01-IM6-84, ... (11 total)
930	955	25	4	30	0	14	0.0	0.1260565809901673	0.8994344763412977	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0041371-T01-IM6-69

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
932	1032	100	6	28	1	13	0.23076923076923078	0.36363115880389973	0.4393389100573004	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53, ... (6 total)
938	988	50	5	29	0	14	0.0	0.06291328848254843	1.2012576136148874	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, EVT-P-0041371-T01-IM6-69
946	1146	200	9	25	2	12	0.30952380952380953	0.25020224491859644	0.6017087979544099	EVT-P-0014403-T01-IM6-87, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0036357-T01-IM6-84, EVT-P-0037167-T01-IM6-81, ... (9 total)
949	974	25	4	30	0	14	0.0	0.1260565809901673	0.8994344763412977	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0041371-T01-IM6-69
949	999	50	5	29	0	14	0.0	0.06291328848254843	1.2012576136148874	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, EVT-P-0041371-T01-IM6-69
949	1049	100	6	28	1	13	0.23076923076923078	0.36363115880389973	0.4393389100573004	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53, ... (6 total)
949	1149	200	9	25	2	12	0.30952380952380953	0.25020224491859644	0.6017087979544099	EVT-P-0014403-T01-IM6-87, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0036357-T01-IM6-84, EVT-P-0037167-T01-IM6-81, ... (9 total)
955	1155	200	6	28	2	12	0.6666666666666666	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53, ... (6 total)
959	1159	200	6	28	2	12	0.6666666666666666	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53, EVT-P-0043910-T01-IM6-89, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
988	1188	200	6	28	2	12	0.6666666666666666	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53, EVT-P-0043910-T01-IM6-89, ... (6 total)
1008	1208	200	6	28	2	12	0.6666666666666666	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0038990-T01-IM6-53, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, ... (6 total)
1032	1232	200	6	28	2	12	0.6666666666666666	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0038990-T01-IM6-53, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, ... (6 total)
1054	1254	200	7	27	3	11	1.0909090909090908	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, ... (7 total)
1059	1159	100	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0043910-T01-IM6-89, EVT-P-0057341-T01-IM6-73
1090	1190	100	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0043910-T01-IM6-89, EVT-P-0057341-T01-IM6-73
1090	1290	200	7	27	3	11	1.0909090909090908	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, ... (7 total)
1108	1208	100	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0014403-T01-IM6-87, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0057341-T01-IM6-73
1121	1221	100	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0014403-T01-IM6-87, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0057341-T01-IM6-73
1121	1321	200	6	28	3	11	1.5454545454545454	0.6721736272626151	0.1725185310795818	EVT-P-0014403-T01-IM6-87, EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0057341-T01-IM6-73, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1146	1346	200	5	29	3	11	2.4545454545454546	0.6271778506123857	0.2026092875791582	EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0057341-T01-IM6-73, EVT-P-0060111-T01-IM7-58
1154	1254	100	4	30	2	12	1.5	1.0	-0.0	EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0060111-T01-IM7-58
1155	1355	200	5	29	3	11	2.4545454545454546	0.6271778506123857	0.2026092875791582	EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0060111-T01-IM7-58
1159	1259	100	4	30	2	12	1.5	1.0	-0.0	EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0060111-T01-IM7-58
1159	1359	200	5	29	3	11	2.4545454545454546	0.6271778506123857	0.2026092875791582	EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0060111-T01-IM7-58
1178	1378	200	5	29	3	11	2.4545454545454546	0.6271778506123857	0.2026092875791582	EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0048892-T01-IM6-90, EVT-P-0060015-T01-IM7-92, EVT-P-0060111-T01-IM7-58
1181	1381	200	6	28	3	11	1.5454545454545454	0.6721736272626151	0.1725185310795818	EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0048892-T01-IM6-90, EVT-P-0060015-T01-IM7-92, ... (6 total)
1191	1391	200	7	27	3	11	1.0909090909090908	1.0	-0.0	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0048892-T01-IM6-90, ... (7 total)
1208	1408	200	7	27	3	11	1.0909090909090908	1.0	-0.0	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0048892-T01-IM6-90, ... (7 total)
1254	1454	200	6	28	3	11	1.5454545454545454	0.6721736272626151	0.1725185310795818	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0060015-T01-IM7-92, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1291	1391	100	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0060015-T01-IM7-92
1341	1391	50	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0060015-T01-IM7-92
1355	1405	50	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0060015-T01-IM7-92
1355	1455	100	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0060015-T01-IM7-92
1355	1555	200	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0060015-T01-IM7-92

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
788	988	200	11	23	0	14	0.0	0.0006583204286534785	3.1815626679007503	EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, ... (11 total)
749	949	200	10	24	0	14	0.0	0.0017219568583829524	2.7639777334868523	EVT-P-0002174-T01-IM3-55, EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, ... (10 total)
755	955	200	10	24	0	14	0.0	0.0017219568583829524	2.7639777334868523	EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, ... (10 total)
855	955	100	9	25	0	14	0.0	0.004385443124565024	2.3579865170458922	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (9 total)
888	988	100	9	25	0	14	0.0	0.004385443124565024	2.3579865170458922	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0024871-T01-IM6-7, EVT-P-0037167-T01-IM6-81, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
849	949	100	8	26	0	14	0.0	0.010884048229464706	1.963209542439507	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (8 total)
923	1023	100	8	26	0	14	0.0	0.010884048229464706	1.963209542439507	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, ... (8 total)
832	1032	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (11 total)
890	1090	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0024871-T01-IM6-7, EVT-P-0036357-T01-IM6-84, ... (11 total)
921	1121	200	11	23	1	13	0.07692307692307693	0.011028775618515856	1.9574726989503282	EVT-P-0005530-T02-IM5-98, EVT-P-0014403-T01-IM6-87, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0036357-T01-IM6-84, ... (11 total)
594	644	50	4	30	4	10	None	0.021584440227703984	1.6658592098833966	EVT-P-0003762-T01-IM5-5, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94
905	955	50	7	27	0	14	0.0	0.026212279835586775	1.5814952042744805	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0041371-T01-IM6-69, ... (7 total)
723	923	200	7	27	0	14	0.0	0.026212279835586775	1.5814952042744805	EVT-P-0002174-T01-IM3-55, EVT-P-0005530-T02-IM5-98, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (7 total)
899	949	50	6	28	0	14	0.0	0.031052774026993746	1.5078995970718483	EVT-P-0005530-T02-IM5-98, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0041371-T01-IM6-69, EVT-P-0043706-T01-IM6-82, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
603	703	100	6	28	5	9	10.555555555555555	0.06082441572037855	1.215922054211928	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94, ... (6 total)
938	988	50	5	29	0	14	0.0	0.06291328848254843	1.2012576136148874	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, EVT-P-0041371-T01-IM6-69
823	923	100	5	29	0	14	0.0	0.06291328848254843	1.2012576136148874	EVT-P-0005530-T02-IM5-98, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, EVT-P-0043706-T01-IM6-82, EVT-P-0046675-T01-IM6-88
550	750	200	7	27	5	9	5.0	0.09691992885737566	1.013586913321784	EVT-P-0002174-T01-IM3-55, EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, ... (7 total)
930	955	25	4	30	0	14	0.0	0.1260565809901673	0.8994344763412977	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0041371-T01-IM6-69
873	923	50	4	30	0	14	0.0	0.1260565809901673	0.8994344763412977	EVT-P-0005530-T02-IM5-98, EVT-P-0024871-T01-IM6-7, EVT-P-0043706-T01-IM6-82, EVT-P-0046675-T01-IM6-88
600	700	100	5	29	4	10	7.6	0.13486142257489503	0.8701122635147213	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0048104-T01-IM6-94
637	737	100	5	29	4	10	7.6	0.13486142257489503	0.8701122635147213	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0039125-T01-IM6-1, EVT-P-0046790-T01-IM6-83, EVT-P-0065545-T01-IM7-95
946	1146	200	9	25	2	12	0.30952380952380953	0.25020224491859644	0.6017087979544099	EVT-P-0014403-T01-IM6-87, EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0036357-T01-IM6-84, EVT-P-0037167-T01-IM6-81, ... (9 total)
644	744	100	4	30	3	11	5.181818181818182	0.28303432810074175	0.5481608874418109	EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0046790-T01-IM6-83, EVT-P-0065545-T01-IM7-95

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
932	1032	100	6	28	1	13	0.23076923076923078	0.36363115880389973	0.4393389100573004	EVT-P-0016507-T01-IM6-74, EVT-P-0019776-T01-IM6-100, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53, ... (6 total)
697	897	200	6	28	1	13	0.23076923076923078	0.36363115880389973	0.4393389100573004	EVT-P-0002174-T01-IM3-55, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, ... (6 total)
670	870	200	5	29	1	13	0.3076923076923077	0.3786297510206428	0.42178526412695666	EVT-P-0002174-T01-IM3-55, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0065545-T01-IM7-95
703	903	200	5	29	1	13	0.3076923076923077	0.3786297510206428	0.42178526412695666	EVT-P-0002174-T01-IM3-55, EVT-P-0009835-T01-IM5-67, EVT-P-0023516-T01-IM6-78, EVT-P-0024871-T01-IM6-7, EVT-P-0065545-T01-IM7-95
607	807	200	7	27	4	10	2.2666666666666666	0.40991252907270703	0.3873088071678072	EVT-P-0002174-T01-IM3-55, EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0039125-T01-IM6-1, ... (7 total)
1341	1391	50	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0060015-T01-IM7-92
1059	1159	100	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0043910-T01-IM6-89, EVT-P-0057341-T01-IM6-73
1108	1208	100	4	30	1	13	0.4358974358974359	0.6271778506123857	0.2026092875791582	EVT-P-0014403-T01-IM6-87, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0057341-T01-IM6-73
1146	1346	200	5	29	3	11	2.4545454545454546	0.6271778506123857	0.2026092875791582	EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0057341-T01-IM6-73, EVT-P-0060111-T01-IM7-58
1155	1355	200	5	29	3	11	2.4545454545454546	0.6271778506123857	0.2026092875791582	EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0060111-T01-IM7-58

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1178	1378	200	5	29	3	11	2.4545454545454546	0.6271778506123857	0.2026092875791582	EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0048892-T01-IM6-90, EVT-P-0060015-T01-IM7-92, EVT-P-0060111-T01-IM7-58
644	844	200	6	28	3	11	1.5454545454545454	0.6721736272626151	0.1725185310795818	EVT-P-0002174-T01-IM3-55, EVT-P-0003762-T01-IM5-5, EVT-P-0008960-T01-IM5-56, EVT-P-0009835-T01-IM5-67, EVT-P-0046790-T01-IM6-83, ... (6 total)
1121	1321	200	6	28	3	11	1.5454545454545454	0.6721736272626151	0.1725185310795818	EVT-P-0014403-T01-IM6-87, EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0057341-T01-IM6-73, ... (6 total)
1181	1381	200	6	28	3	11	1.5454545454545454	0.6721736272626151	0.1725185310795818	EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0048892-T01-IM6-90, EVT-P-0060015-T01-IM7-92, ... (6 total)
1254	1454	200	6	28	3	11	1.5454545454545454	0.6721736272626151	0.1725185310795818	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0060015-T01-IM7-92, ... (6 total)
1154	1254	100	4	30	2	12	1.5	1.0	-0.0	EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, EVT-P-0060111-T01-IM7-58
955	1155	200	6	28	2	12	0.6666666666666666	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0037167-T01-IM6-81, EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53, ... (6 total)
959	1159	200	6	28	2	12	0.6666666666666666	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0038406-T01-IM6-80, EVT-P-0038990-T01-IM6-53, EVT-P-0043910-T01-IM6-89, ... (6 total)
1008	1208	200	6	28	2	12	0.6666666666666666	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0038990-T01-IM6-53, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1054	1254	200	7	27	3	11	1.0909090909090908	1.0	-0.0	EVT-P-0014403-T01-IM6-87, EVT-P-0036357-T01-IM6-84, EVT-P-0043326-T01-IM6-71, EVT-P-0043910-T01-IM6-89, EVT-P-0048892-T01-IM6-90, ... (7 total)
1191	1391	200	7	27	3	11	1.0909090909090908	1.0	-0.0	EVT-P-0023550-T01-IM6-68, EVT-P-0026288-T03-IM6-91, EVT-P-0040752-T01-IM6-62, EVT-P-0043326-T01-IM6-71, EVT-P-0048892-T01-IM6-90, ... (7 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039125-T01-IM6-1	P-0039125-T01-IM6	P-0039125	Breast Cancer	IDC	AKAP8L-CDK12	AKAP8L	unknown	three_prime	37627994	2	637	False	637	True	exact match	retained	1.0	False	Protein kinase domain			AKAP8L (NM_014371) - CDK12 (NM_016507) rearrangement: t(17;19)(q12;p13.12)(chr17:g.37627994::chr19:g.15529663) | Protein Fusion: mid-exon {AKAP8L:CDK12}	NA
EVT-P-0056622-T01-IM6-2	P-0056622-T01-IM6	P-0056622	Breast Cancer	IDC	ANKFN1-CDK12	ANKFN1	in-frame	three_prime	37619399	2	349	True	349	True	exact match	retained	1.0	False	Protein kinase domain			ANKFN1 (NM_153228) - CDK12 (NM_016507) rearrangement: c.62+8631:ANKFN1_c.1046+29:CDK12dup | Protein Fusion: in frame {ANKFN1:CDK12}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003762-T02-IM5-4	P-0003762-T02-IM5	P-0003762	Breast Cancer	IDC	CDK12-ADCY9	ADCY9	unknown	three_prime	37646665	3	644	True	644	True	exact match	retained	1.0	False	Protein kinase domain			ADCY9 (NM_001116) - CDK12 (NM_016507) rearrangement: t(16;17)(p13.3;q12)(chr16:g.40333430::chr17:g.37646665) | Protein fusion: mid-exon (ADCY9-CDK12)	NA
EVT-P-0003762-T01-IM5-5	P-0003762-T01-IM5	P-0003762	Breast Cancer	IDC	CDK12-ADCY9	ADCY9	unknown	three_prime	37646665	3	644	True	644	True	exact match	retained	1.0	False	Protein kinase domain			t(16;17)(p13.3;q12)(chr16:g.40333430::chr17:g.37646665) | Protein fusion: mid-exon (ADCY9-CDK12)	NA
EVT-P-0024871-T01-IM6-7	P-0024871-T01-IM6	P-0024871	Breast Cancer	IDC	CDK12-CACNA1G	CACNA1G	unknown	five_prime	37667806	8	897	False	897	True	exact match	disrupted	0.5802047781569966	True			Protein kinase domain	CDK12 (NM_016507) - CACNA1G (NM_018896) rearrangement: c.2691:CDK12_c.6399+226:CACNA1Gdel | Protein Fusion: mid-exon {CDK12:CACNA1G}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038990-T01-IM6-53	P-0038990-T01-IM6	P-0038990	Colorectal Cancer	READ	CDK12-FBXL20	FBXL20	in-frame	five_prime	37676367	11	1032	True	1031	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - FBXL20 (NM_032875) rearrangement: c.3095+27:CDK12_c.104+18457:FBXL20inv | Protein Fusion: in frame {CDK12:FBXL20}	NA
EVT-P-0002174-T01-IM3-55	P-0002174-T01-IM3	P-0002174	Endometrial Cancer	UMEC	CDK12-GRB7	GRB7	unknown	three_prime	37650768	5	750	True	750	True	exact match	disrupted	0.924914675767 9181	True			Protein kinase domain	NA | Transcript fusion (GRB7-CDK12)	NA
EVT-P-0008960-T01-IM5-56	P-0008960-T01-IM5	P-0008960	Prostate Cancer	PRAD	CDK12-HTR1E	HTR1E	unknown	five_prime	37646977	3	700	False	699	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697112::chr17:g.37646977) | Protein fusion: mid-exon (CDK12-HTR1E)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008960-T02-IM6-57	P-0008960-T02-IM6	P-0008960	Prostate Cancer	PRAD	CDK12-HTR1E	HTR1E	unknown	five_prime	37646977	3	700	False	699	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697113::chr17:g.37646977) | Protein Fusion: mid-exon {CDK12:HTR1E}	NA
EVT-P-0060111-T01-IM7-58	P-0060111-T01-IM7	P-0060111	Anal Cancer	ANSC	CDK12-IKZF3	IKZF3	in-frame	five_prime	37682742	13	1254	True	1253	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - IKZF3 (NM_012481) rearrangement: c.3760+173:CDK12_c.8-12056:IKZF3inv | Protein Fusion: in frame {CDK12:IKZF3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040752-T01-IM6-62	P-0040752-T01-IM6	P-0040752	Endometrial Cancer	USC	CDK12-MTRES1	MTRES1	unknown	five_prime	37687160	14	1355	False	1354	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - C6orf203 (NM_016487) rearrangement: t(6;17)(q21;q12)(chr6:g.107368450::chr17:g.37687160) | Protein Fusion: mid-exon {CDK12:C6orf203}	NA
EVT-P-0009835-T01-IM5-67	P-0009835-T01-IM5	P-0009835	Endometrial Cancer	USC	CDK12-RECQL4	RECQL4	out-of-frame	five_prime	37656296	5	807	True	806	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.2730375426621605	True			Protein kinase domain	CDK12 (NM_016507) - RECQL4 (NM_004260) rearrangement: t(8;17)(q24.3;q12)(chr8:g.145739493::chr17:g.37656296) | Protein fusion: out of frame (CDK12-RECQL4)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023550-T01-IM6-68	P-0023550-T01-IM6	P-0023550	Prostate Cancer	PRAD	CDK12-RPS6KB1	RPS6KB1	unknown	three_prime	37687269	14	1391	False	1391	True	exact match	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) rearrangement: c.4173_chr17:g.58028299dup | Protein Fusion: mid-exon {RPS6KB1:CDK12}	NA
EVT-P-0041371-T01-IM6-69	P-0041371-T01-IM6	P-0041371	Breast Cancer	IDC	CDK12-SKAP1	SKAP1	out-of-frame	five_prime	37673680	9	949	True	948	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.757679180887372	True			Protein kinase domain	CDK12 (NM_016507) - SKAP1 (NM_003726) rearrangement: c.2847-13:CDK12_c.280+66555:SKAP1inv | Protein Fusion: out of frame {CDK12:SKAP1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043326-T01-IM6-71	P-0043326-T01-IM6	P-0043326	Breast Cancer	IDC	CDK12-SNF8	SNF8	out-of-frame	five_prime	37683411	13	1254	True	1253	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - SNF8 (NM_007241) rearrangement: c.3760+842:CDK12_c.422+1405:SNF8inv | Protein Fusion: out of frame {CDK12:SNF8}	NA
EVT-P-0057341-T01-IM6-73	P-0057341-T01-IM6	P-0057341	Breast Cancer	IDC	CDK12-SPON1	SPON1	unknown	five_prime	37682247	13	1146	False	1146	True	exact match	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - SPON1 (NM_006108) rearrangement: t(11;17)(p15.2;q12)(chr11:g.14149386::chr17:g.37682247) | Protein Fusion: mid-exon {CDK12:SPON1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016507-T01-IM6-74	P-0016507-T01-IM6	P-0016507	Salivary Gland Cancer	HNMASC	CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	10	949	True	949	True	exact match	disrupted	0.2457337883959043	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel | Antisense Fusion	NA
EVT-P-0016507-T02-IM6-75	P-0016507-T02-IM6	P-0016507	Salivary Gland Cancer	HNMASC	CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	10	949	True	949	True	exact match	disrupted	0.2457337883959043	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel | Antisense Fusion	NA
EVT-P-0016507-T03-IM6-76	P-0016507-T03-IM6	P-0016507	Salivary Gland Cancer	HNMASC	CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	10	949	True	949	True	exact match	disrupted	0.2457337883959043	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel. | Antisense Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016507-T04-IM6-77	P-0016507-T04-IM6	P-0016507	Salivary Gland Cancer	HNMASC	CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	10	949	True	949	True	exact match	disrupted	0.2457337883959043	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel | Antisense Fusion	NA
EVT-P-0023516-T01-IM6-78	P-0023516-T01-IM6	P-0023516	Colorectal Cancer	COAD	COLEC12-CDK12	COLEC12	out-of-frame	three_prime	37665943	7	870	True	870	True	exact match	disrupted	0.515358361774744	True			Protein kinase domain	COLEC12 (NM_130386) - CDK12(NM_016507) rearrangement: t(17;18)(q12;p11.32)(chr17:g.37665943::chr18:g.386902) | Protein Fusion: out of frame {COLEC12:CDK12}	NA
EVT-P-0038406-T01-IM6-80	P-0038406-T01-IM6	P-0038406	Ovarian Cancer	CCOV	CSF3-CDK12	CSF3	out-of-frame	three_prime	37676083	11	988	True	988	True	exact match	disrupted	0.11262798634812286	True			Protein kinase domain	CSF3 (NM_000759) - CDK12 (NM_016507) Rearrangement : c.313-58:CSF3_c.2964-126:CDK12dup | Protein Fusion: out of frame {CSF3:CDK12}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037167-T01-IM6-81	P-0037167-T01-IM6	P-0037167	Ovarian Cancer	HGSOC	ERBB2-CDK12	ERBB2	unknown	three_prime	37673710	10	955	False	955	True	exact match	disrupted	0.2252559726962 4573	True			Protein kinase domain	ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.2553:ERBB2_c.2864:CDK12 dup | Protein Fusion: mid-exon {ERBB2:CDK12}	NA
EVT-P-0043706-T01-IM6-82	P-0043706-T01-IM6	P-0043706	Ovarian Cancer	HGSOC	ERBB2-CDK12	ERBB2	unknown	three_prime	37668120	9	923	True	923	True	exact match	disrupted	0.3344709897610 9213	True			Protein kinase domain	ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.2293:ERBB2_c.2768+237:CDK12dup | Protein Fusion: mid-exon {ERBB2:CDK12}	NA
EVT-P-0046790-T01-IM6-83	P-0046790-T01-IM6	P-0046790	Hepatobiliary Cancer	GBC	ERBB2-CDK12	ERBB2	out-of-frame	three_prime	37640037	3	644	True	644	True	exact match	retained	1.0	False	Protein kinase domain			ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.225+207:ERBB2_c.1932-6773:CDK12 dup | Protein Fusion: out of frame {ERBB2:CDK12}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036357-T01-IM6-84	P-0036357-T01-IM6	P-0036357	Bladder Cancer	BLCA	FAM222B-CDK12	FAM222B	unknown	three_prime	37681100	12	1090	False	1090	True	exact match	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) rearrangement: c.3269:CDK12_chr17:g.27096697inv | Protein Fusion: mid-exon {FAM222B:CDK12}	NA
EVT-P-0036553-T01-IM6-85	P-0036553-T01-IM6	P-0036553	Colorectal Cancer	READ	FBXL20-CDK12	FBXL20	unknown	three_prime	37619062	1	246	False	246	True	exact match	retained	1.0	False	Protein kinase domain			FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv | Protein Fusion: mid-exon {FBXL20:CDK12}	NA
EVT-P-0014403-T01-IM6-87	P-0014403-T01-IM6	P-0014403	Prostate Cancer	PRAD	FBXL20-CDK12	FBXL20	unknown	three_prime	37682171	13	1121	False	1121	True	exact match	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.43-9338:FBXL20_c.3362:CDK12inv | Protein fusion: mid-exon (FBXL20-CDK12)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0046675-T01-IM6-88	P-0046675-T01-IM6	P-0046675	Cancer of Unknown Primary	SCCNOS	FBXL20-CDK12	FBXL20	in-frame	three_prime	37671717	9	923	True	923	True	exact match	disrupted	0.3344709897610921 3	True			Protein kinase domain	FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.398+4667:FBXL20_c.2769-267:CDK12inv | Protein Fusion: in frame {FBXL20:CDK12}	NA
EVT-P-0043910-T01-IM6-89	P-0043910-T01-IM6	P-0043910	Head and Neck Cancer	SNUC	FBXL20-CDK12	FBXL20	unknown	three_prime	37682285	13	1159	False	1159	True	exact match	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv | Protein Fusion: mid-exon {FBXL20:CDK12}	NA
EVT-P-0048892-T01-IM6-90	P-0048892-T01-IM6	P-0048892	Breast Cancer	IDC	FBXL20-CDK12	FBXL20	unknown	three_prime	37682432	13	1208	False	1208	True	exact match	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv | Protein Fusion: mid-exon {FBXL20:CDK12}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026288-T03-IM6-91	P-0026288-T03-IM6	P-0026288	Breast Cancer	BRCA	FGFR4-CDK12	FGFR4	unknown	three_prime	37687238	14	1381	False	1381	True	exact match	lost	0.0	False		Protein kinase domain		FGFR4 (NM_213647) - CDK12 (NM_016507) rearrangement: t(5,17)(q35.2;q12)(chr5:g.176518227::chr17:g.37687238) | Protein Fusion: mid-exon {FGFR4:CDK12}	NA
EVT-P-0060015-T01-IM7-92	P-0060015-T01-IM7	P-0060015	Colorectal Cancer	COAD	IGFBP4-CDK12	IGFBP4	unknown	three_prime	37687228	14	1378	False	1378	True	exact match	lost	0.0	False		Protein kinase domain		IGFBP4 (NM_001552) - CDK12 (NM_016507) rearrangement: c.349+3777:IGFBP4_c.4132:CDK12dup | Protein Fusion: mid-exon {IGFBP4:CDK12}	NA
EVT-P-0068695-T01-IM7-93	P-0068695-T01-IM7	P-0068695	Cancer of Unknown Primary	CUP	IKZF3-CDK12	IKZF3	unknown	three_prime	37619043	1	240	False	240	True	exact match	retained	1.0	False	Protein kinase domain			IKZF3 (NM_012481) - CDK12 (NM_016507) rearrangement: c.8-3620:IKZF3_c.719:CDK12inv | Protein Fusion: mid-exon {IKZF3:CDK12}	NA

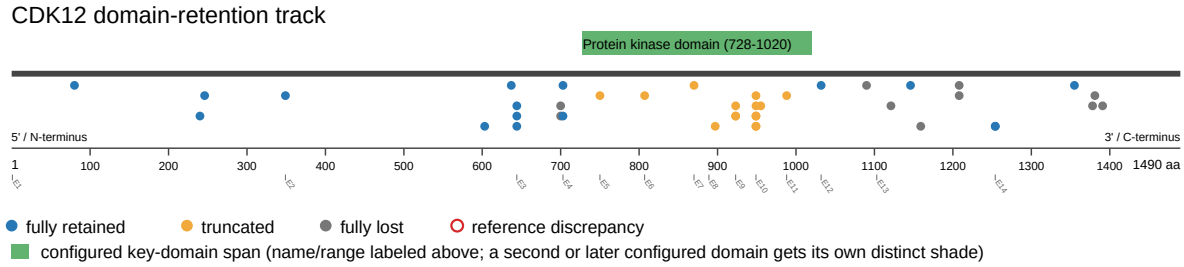
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048104-T01-IM6-94	P-0048104-T01-IM6	P-0048104	Breast Cancer	BRCA	MED13-CDK12	MED13	unknown	three_prime	37627894	2	603	False	603	True	exact match	retained	1.0	False	Protein kinase domain			MED13 (NM_005121) - CDK12 (NM_016507) rearrangement: c.2264-2554:MED13_c.1809:CDK12inv | Protein Fusion: mid-exon {MED13:CDK12}	NA
EVT-P-0065545-T01-IM7-95	P-0065545-T01-IM7	P-0065545	Esophagogastric Cancer	STAD	ODAD4-CDK12	ODAD4	in-frame	three_prime	37648612	4	703	True	703	True	exact match	retained	1.0	False	Protein kinase domain			TTC25 (NM_031421) - CDK12 (NM_016507) rearrangement: c.1059-1499:TTC25_c.2109-392:CDK12dup | Protein Fusion: in frame {TTC25:CDK12}	NA
EVT-P-0048892-T01-IM6-97	P-0048892-T01-IM6	P-0048892	Breast Cancer	IDC	SEC31A-CDK12	SEC31A	unknown	three_prime	37682432	13	1208	False	1208	True	exact match	lost	0.0	False		Protein kinase domain		SEC31A (NM_014933) - CDK12 (NM_016507) rearrangement: t(4;17)(q21.22;q12)(chr4:g.83761890::chr17:g.37682432) | Protein Fusion: mid-exon {SEC31A:CDK12}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005530-T02-IM5-98	P-0005530-T02-IM5	P-0005530	Cancer of Unknown Primary	CUP	SYNRG-CDK12	SYNRG	out-of-frame	three_prime	37671800	9	923	True	923	True	exact match	disrupted	0.3344709897610213	True			Protein kinase domain	SYNRG (NM_007247) - CDK12 (NM_016507) rearrangement : c.118+961:SYNRG_c.2769-184:CDK12 inv | Protein fusion: out of frame (SYNRG-CDK12)	NA
EVT-P-0041371-T01-IM6-99	P-0041371-T01-IM6	P-0041371	Breast Cancer	IDC	TBKBP1-CDK12	TBKBP1	in-frame	three_prime	37648975	4	703	True	703	True	exact match	retained	1.0	False	Protein kinase domain			TBKBP1 (NM_014726) - CDK12 (NM_016507) rearrangement: c.872+1376:TBKBP1_c.2109-29:CDK12 dup | Protein Fusion: in frame {TBKBP1:CDK12}	NA
EVT-P-0019776-T01-IM6-100	P-0019776-T01-IM6	P-0019776	Breast Cancer	IDC	USP6-CDK12	USP6	in-frame	three_prime	37673592	10	949	True	949	True	exact match	disrupted	0.2457337883959043	True			Protein kinase domain	USP6 (NM_004505) - CDK12 (NM_016507) rearrangement: c.2916-470:USP6_c.2847-101:CDK12del | Protein Fusion: in frame {USP6:CDK12}	NA

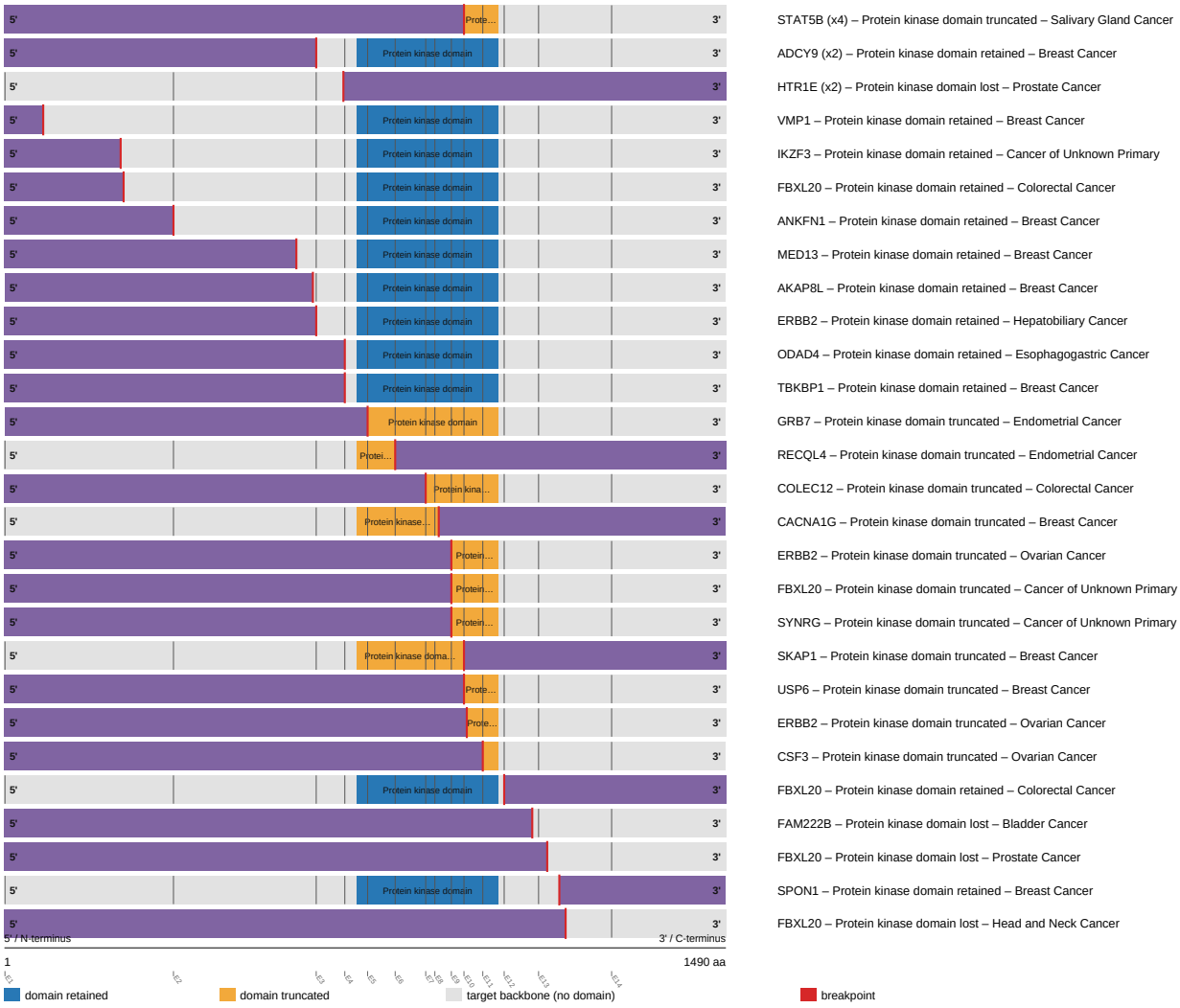
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059578-T01-IM7-101	P-0059578-T01-IM7	P-0059578	Breast Cancer	IDC	VMP1-CDK12	VMP1	unknown	three_prime	37618564	1	80	False	80	True	exact match	retained	1.0	False	Protein kinase domain			VMP1 (NM_030938) - CDK12 (NM_016507) rearrangement: c.582+1747:VMP1_c.240:CDK12dup | Protein Fusion: mid-exon {VMP1:CDK12}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 36 partner/breakpoint groups by recurrence; see results.tsv for the complete list.